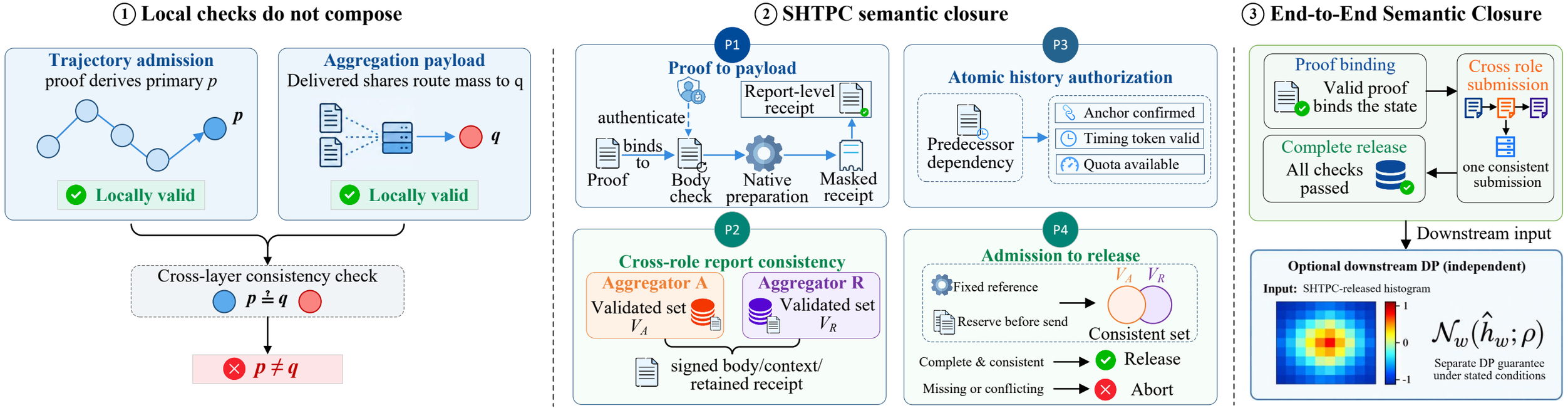

① Local checks do not compose
Trajectory admission
proof derives primary p
p
Locally valid
Aggregation payload
Delivered shares route mass to q
q
Locally valid
Cross-layer consistency check
p ≟ q
p ≠ q
② SHTPC semantic closure
P1
Proof to payload
authenticate
Report-level receipt
Proof
binds to
Body check
Native preparation
Masked receipt
P3
Atomic history authorization
Predecessor dependency
Anchor confirmed
Timing token valid
Quota available
P2
Cross-role report consistency
Aggregator A
Validated set V_A
Aggregator R
Validated set V_R
signed body/context/ retained receipt
P4
Admission to release
Fixed reference
Reserve before send
V_A V_R
Consistent set
Complete & consistent
Release
Missing or conflicting
Abort
③ End-to-End Semantic Closure
Proof binding
Valid proof binds the state
Cross role submission
Complete release
All checks passed
one consistent submission
Downstream input
Optional downstream DP (independent)
Input: SHTPC-released histogram
1
0
-1
$\mathcal{N}_w(\hat{h}_w;\rho)$
Separate DP guarantee under stated conditions